



Polynomial Time Algorithms for Hop Domination

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Abstract. A set $S \subseteq V(G)$ is said to be a hop dominating set if every vertex $u \in V(G) \setminus S$, there exists a vertex $v \in S$ such that $d(u, v) = 2$ where $d(u, v)$ represents the distance between u and v in G . The minimum k for which there exists a hop dominating set of size k is called the *hop domination number* denoted by $\gamma_h(G)$. Henning et al. (Inf. Process. Lett. 2020) showed that HOP DOMINATION is NP-hard for bipartite graphs and chordal graphs. The following are the results of this paper.

- Henning et al. presented a linear time algorithm for solving HOP DOMINATION on bipartite permutation graphs which is a proper subset of biconvex bipartite graphs. In this paper we present a polynomial algorithm for the problem on biconvex bipartite graphs, a superclass of bipartite permutation graphs.
- We show that HOP DOMINATION is polynomial time solvable on interval graphs which are a subclass of chordal graphs.
- We initiate the study on this problem from the parameterized complexity perspective. We show that the decision version of HOP DOMINATION is W[1]-hard when parameterized by solution size.

Keywords: domination · hop domination · interval graphs · biconvex bipartite graphs

1 Introduction

Given a graph $G = (V, E)$, a set $D \subseteq V(G)$ is said to be a *dominating set* if for every vertex v in G , either $v \in D$ or v has at least one neighbor in D . The minimum cardinality of a dominating set is called the *domination number* of G and is denoted by $\gamma(G)$. Two vertices u and v in a graph G are said to *k-step dominate* each other if $d_G(u, v) = k$, where $k \geq 1$ is an integer. A set $S \subseteq V(G)$ is called a *k-step dominating set* of G if every vertex of G is *k-step dominated* by some vertex in S . The *k-step dominating set* problem asks to find a minimum size of a *k-step dominating set*. Chartrand et al. (1995) introduced and studied 2-step domination in graphs [1]. A generalization of the *k-step domination* problem

called the *step domination* problem has been studied where the input is a graph G , and integer k_i associated with each vertex $v_i \in V(G)$. The objective is to find a set $S \subseteq V(G)$ of minimum size such that $N_{k_i}(S)$ forms a partition of $V(G)$, where $N_k(v)$ denotes the set of vertices at distance k from v . Hersh [2] generalized to n steps the notion of exact 2-step domination introduced by Chartrand et al. [1].

A set $S \subseteq V(G)$ is said to be a *hop dominating set* if every vertex $u \in V(G) \setminus S$, there exists a vertex $v \in S$ such that u is 2-step dominated by v . Given a graph G , the HOP DOMINATION problem asks to find the minimum k for which there exists a hop dominating set of size k , called the *hop domination number* denoted by $\gamma_h(G)$. Several bounds on this problem have been studied [3–5].

Henning et al. [5] presented upper bounds for the hop domination number of a graph. Further, they showed that the decision version of HOP DOMINATION is NP-complete for planar bipartite graphs and planar chordal graphs. Henning et al. [6] presented a linear time algorithm to compute the minimum hop domination number in bipartite permutation graphs. Further, they showed that the decision version of HOP DOMINATION is NP-complete for perfect elimination graphs. Also, they showed that the hop domination number could not be approximated within a factor of $(1 - \epsilon) \log |V|$, unless $P=NP$ and gave an approximation algorithm with a factor of $1 + \log(\Delta(\Delta - 1) + 1)$, where Δ is the maximum degree in G . Further, they showed that HOP DOMINATION is APX-complete for bipartite graphs of maximum degree 3.

Our results and Discussion. In this paper we extend the study on HOP DOMINATION. Henning et al. [6] showed that HOP DOMINATION is NP-hard on chordal graphs and bipartite graphs. We show that HOP DOMINATION is polynomial time solvable on interval graphs. The main ingredient of the algorithm is the structural property admitted by interval graphs called “multi-chain ordering”. This result is presented in Sect. 4. Henning et al. [6] presented a linear time algorithm to compute the minimum hop domination number in bipartite permutation graphs. We consider biconvex bipartite graphs a superclass of bipartite permutation graphs and give a polynomial time algorithm. This result is presented in Sect. 5.

We initiate the study of parameterized complexity on this problem. We show that HOP DOMINATION is $W[1]$ -hard parameterized by solution size and this result is presented in Sect. 3.

As open questions and future directions, one could explore the parameterized complexity of the problem on graphs of bounded treewidth, modular width, distance to cluster and cliquewidth. Another direction could be to obtain polynomial kernels for the problem with respect to various structural parameters. It will also be interesting to find the complexity of the problem on split graphs, distance hereditary graphs, etc.

2 Preliminaries

Let $G = (V, E)$ be a graph with vertex set $V = V(G)$ and edge set $E = E(G)$. A set of vertices $I \subseteq V(G)$ is said to be *independent* if no two vertices in I are adjacent to each other. A *bipartite graph* $G = (X, Y, E)$ is a graph in which the vertex set $V(G)$ can be partitioned into two sets X and Y such that every edge connects a vertex in X to a vertex in Y . The distance between two vertices u and v in a graph G is denoted by $d(u, v)$ which is the number of edges on a shortest path between them. The diameter $\text{diam}(G)$ of a graph G is the maximum distance over all pairs of vertices $u, v \in V(G)$. We denote the set $\{1, 2, \dots, p\}$ with $[p]$ and the set $\{a, a+1, \dots, b\}$ with $[a, b]$. A vertex $v \in V(G)$ is said to be *universal* if for every $u \in V(G) \setminus \{v\}$, $uv \in E(G)$.

In parameterized complexity, in addition to the input we are given a parameter k and the objective is to obtain an algorithm running in time $f(k)n^{O(1)}$ where f is some computable function dependent only on k and n is the size of the input. We say a problem belongs to the class *fixed parameter tractable* if there exists an algorithm running in above mentioned time. Otherwise, the problem belongs to the class *W-hard*. For more details the reader is directed to [7]. The proofs of the results marked $(\star)$ are presented in the full version of the paper. We now look at the following observation, the proof of which is trivial.

Observation 1. *If u is universal in G , then any hop dominating set contains u .*

3 W[1]-Hard Parameterized by Solution Size

In this section, we initiate the study of the problem from a parameterized complexity perspective. As a first step, we study the problem with respect to the natural parameter, the solution size. We show that HOP DOMINATION is W[1]-hard when parameterized by solution size. Towards this, we show a reduction from MULTICOLORED INDEPENDENT SET which takes as input a graph G , an integer k , and a partition of $V(G)$ into k sets $V_1, V_2, \dots, V_k$ and the objective is to find an independent set $S \subseteq V(G)$ of size k such that $|S \cap V_i| = 1$ for each i .

Theorem 1. *HOP DOMINATION is W[1]-hard parameterized by solution size.*

Proof. We give a reduction from MULTICOLORED INDEPENDENT SET to HOP DOMINATION. Let $G = (V, E)$ and a partition $V_1, V_2, \dots, V_k$ of $V(G)$ be an instance of MULTICOLORED INDEPENDENT SET. We assume that $k \geq 3$ and each V_i is an independent set, for all i . We construct a graph G' from G as follows. An illustration of the gadget is given in Fig. 1.

1. Add a copy of $V_1, V_2, \dots, V_k$ to G' .
2. For each pair i and j , $i \neq j$, each vertex in V_i is adjacent to every vertex in V_j .
3. For each i , add a vertex u_i and make it adjacent to all vertices in V_i .

4. For each i , add two vertices u_i^1 and u_i^2 , and add the edges $u_i u_i^1$, $u_i u_i^2$ and $u_i^1 u_i^2$.
5. For each pair of vertices $u \in V_i$ and $v \in V_j$ such that $uv \in E(G)$, we add a vertex e_{uv} and make it adjacent to all vertices in $(V_i \cup V_j) \setminus \{u, v\}$.
6. For each vertex e_{uv} , add a vertex e'_{uv} and make it adjacent to e_{uv} .

Claim 1. *Let D be a hop dominating set of G' of size k then $|D \cap \{V_i, u_i, u_i^1, u_i^2\}| = 1$ for all i . Further $|D \cap V_i| = 1$, for all i .*

Proof. Let $A_i = \{V_i, u_i, u_i^1, u_i^2\}$ for all i . Consider a partition V_i . If D hop dominates the vertices u_i^1 and u_i^2 , it is the case that either $\{u_i^1, u_i^2\} \subseteq D$ or there exist $q_i \in D \cap V_i$.

Suppose $\{u_i^1, u_i^2\} \subseteq D$. Since cardinality of D is k , there exists some j such that $|D \cap \{V_j, u_j, u_j^1, u_j^2\}| = \emptyset$. The vertices u_j^1 and u_j^2 are at distance at least three from $V(G) \setminus A_j$ and thus are not hop dominated which is a contradiction. Thus it is the case that $q_i \in D \cap V_i$.

Suppose $|D \cap V_i| > 1$. Since D is of size k , there exist some j such that $V_j \cap D = \emptyset$ and the vertices u_j^1 and u_j^2 are not hop dominated which is a contradiction. $\square$

Claim 2. *Let D be a hop dominating set of G' of size k . Then for each $uv \in E(G)$, $|D \cap \{u, v\}| \leq 1$.*

Proof. From Claim 1, it is clear that $|D \cap V_i| = 1$ for each i . Let $u \in V_i$ and $v \in V_j$. Consider the vertex e_{uv} . Suppose for a contradiction that $\{u, v\} \subseteq D$. Then the vertex e'_{uv} is not hop dominated by any of the vertices in D . $\square$

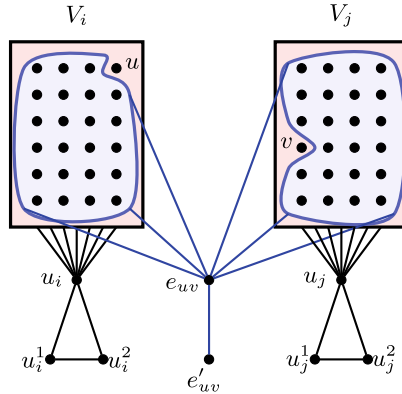


Fig. 1. An illustration of the edges between a pair of sets V_i and V_j . The edges between V_i and V_j are not indicated to avoid clutter.

Lemma 1. *G has a multicolored independent set of size k if and only if G' has a hop dominating set of size k .*

Proof. Suppose G has a multicolored independent set S of size k . Let $S = \{q_1, q_2, \dots, q_k\}$ where $q_i \in V_i$ for all i . We claim that S is a hop dominating set for G' . The vertices u_i^1 and u_i^2 are hop dominated by q_i because u_i is universal to V_i and $q_i \in V_i$. Each vertex u_i is hop dominated by q_j for any $j \neq i$ because $u_i q_i \in E(G')$ and q_i is universal to V_j . For each vertex $w \in V_i \setminus \{q_i\}$, we have that w is hop dominated by q_i because $w u_i \in E(G')$ and u_i is universal to V_i . For each e_{uv} , where $u \in V_i$ and $v \in V_j$, we have that $q_{j'}$ hop dominates e_{uv} where $i \neq j' \neq j$, as e_{uv} is at a distance two from $q_{j'}$. The existence of j' is guaranteed because $k \geq 3$. Now consider the vertex e'_{uv} . Since S is an independent set of G , it is the case that $|S \cap \{u, v\}| \leq 1$. Without loss of generality let $v \notin S$. Then e'_{uv} is hop dominated by q_j because e'_{uv} is adjacent to e_{uv} and e_{uv} is universal to V_j except v , in particular q_j .

We now prove the converse. Suppose D is a hop dominating set for G' of size k . We show that D is a k -multicolored independent set for G . It is sufficient to show that for all pairs of vertices u and v in D either e_{uv} does not exist in G' or $|\{u, v\} \setminus (D \cap \{u, v\})| \geq 1$. The former case indicates that $uv \notin E(G)$ while the latter case indicates that at most one of u and v are in D . If $e_{uv} \notin V(G')$, then we are done. Suppose that $e_{uv} \in V(G')$. From Claims 1 and 2, we get that at most one of u and v are in D and thus D is an independent set in G . $\square$

From Lemma 1, and the fact that MULTICOLORED INDEPENDENT SET is $W[1]$ -complete [7], we obtain our desired result. $\square$

4 Interval Graphs

In this section, we show that HOP DOMINATION can be solved in polynomial time on interval graphs. Before we look at the algorithm, we look at the definition of multi-chain ordering which is crucial to our algorithm.

A bipartite graph $G = (A \cup B, E)$ is called a *chain graph* if for every two vertices $u_1, u_2 \in A$ we have either $N(u_1) \subseteq N(u_2)$ or $N(u_2) \subseteq N(u_1)$. We can see that each part of a chain graph can be linearly ordered under the inclusion of their neighborhoods.

Definition 1 (Multi-chain ordering [8]). *Let $G = (V, E)$ be a connected graph and a partition of $V(G)$ say $L_0, L_1, \dots, L_{q-1}$ be distance layers of $V(G)$ starting from a vertex $v_0 \in V(G)$ where $L_0 = \{v_0\}$. The layer L_i , where $i \in [q-1]$, denotes the set of vertices that are at a distance i from v_0 . Note that q here denotes the largest integer such that L_{q-1} is non-empty. We say that these layers form a multi-chain ordering of G if for every two consecutive layers L_i and L_{i+1} , where $i \in \{0, 1, \dots, q-1\}$, we have that the vertices in L_i and L_{i+1} , and the edges connecting these layers form a chain graph.*

As a consequence of the above definition, we have the following. For each $i \in [q-2]$, since $G[L_i \cup L_{i-1}]$ and $G[L_i \cup L_{i+1}]$ induce a chain graph, there are two orderings one with respect to L_{i-1} and another with respect to L_{i+1} . In the former ordering (resp. latter case), there exists a vertex, say $\alpha_i \in L_i$ (resp. $\beta_i \in L_i$) such that $N(\alpha_i)$ (resp. $N(\beta_i)$) contains every vertex in L_{i-1} (resp. L_{i+1}) that has an edge to a vertex in L_i . Notice that α_i and β_i could be the same vertex. Since L_0 and L_{q-1} are the first and the last layers there exists only one ordering for each of them and thus we β_0 and α_{q-1} respectively. It is known that all connected interval graphs admit multi-chain ordering [8]. Thus we assume that G admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ on q layers that can be computed in polynomial time.

Definition 2 (Interval graph). A graph $G = (V, E)$ is an interval graph if there exists a set $\mathcal{I}$ of closed intervals on the real line such that there is a bijection $f : V \rightarrow \mathcal{I}$ such that $uv \in E(G)$ if and only if $f(u) \cap f(v) \neq \emptyset$.

For each interval $I \in \mathcal{I}$, we use $L(I)$ and $R(I)$ to denote the left and right endpoints of I respectively. Without loss of generality, we can assume that no two intervals share the same endpoint. We designate two special intervals I_ℓ and I_r where $R(I_\ell) < R(I')$, for every $I' \in \mathcal{I} \setminus I_\ell$ and $L(I_r) > L(I')$, for every $I' \in \mathcal{I} \setminus I_r$. That is I_ℓ is the interval with the least right endpoint and I_r is the interval that has the largest left endpoint.

Lemma 2. Let G be an interval graph with $\text{diam}(G)=3$. Then there exists a hop dominating set $S \subseteq V(G)$ such that $|S| \leq 4$.

Proof. Let G be a connected interval graph and $\mathcal{I}$ be an interval representation of G . Recall that $R(I_\ell) < R(I')$, for every $I' \in \mathcal{I} \setminus I_\ell$ and $L(I_r) > L(I')$, for every $I' \in \mathcal{I} \setminus I_r$. By the definition of I_ℓ , I_r and $\text{diam}(G)=3$, we have $d(I_\ell, I_r) = 3$. Then there exist two intervals I_1 and I_2 such that the graph induced on the intervals I_ℓ, I_1, I_2, I_r is a path.

Let **class-a** be the set of intervals in $\mathcal{I}$ that are adjacent to I_ℓ , **class-b** be the set of intervals in $\mathcal{I}$ that are not adjacent to I_ℓ but are adjacent to I_1 , and **class-c** be the set of intervals in $\mathcal{I}$ that are not adjacent to I_ℓ and I_1 . An illustration of the classes is given in Fig. 2. It is easy that any interval other than I_ℓ, I_1, I_2 and I_r belongs to one of these classes.

Claim. Let $S = \{I_\ell, I_1, I_2, I_r\}$. Then S is a hop dominating set.

Proof. Consider an interval $I \in \mathcal{I} \setminus \{S\}$. We now argue that I is hop dominated based on the class it belongs to.

- I belongs to **class-a**.

We first consider when I does not intersect with I_2 . From the definition of I_ℓ and the case assumption, we have that I is adjacent to I_ℓ and I_1 . Thus I is hop dominated by I_2 . Now we consider the case when I intersects with I_2 . Notice that I does not intersect with I_r otherwise $\text{diam}(G)=2$. Therefore I is hop dominated by I_r . Every interval of **class-a** is hop dominated by either I_2 or I_r .

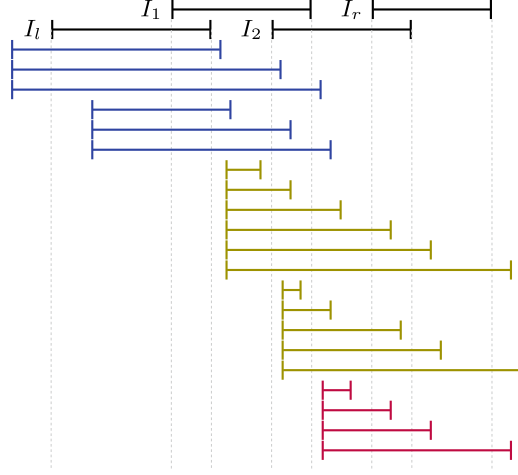


Fig. 2. Interval representation of an interval graph G with $\text{diam}(G) = 3$. The intervals colored in blue, olive and purple denote the intervals of **class-a**, **class-b** and **class-c** respectively.

- I belongs to **class-b**.
From the definition of **class-b** the interval I does not intersect with I_ℓ and intersects I_1 . Therefore I is hop dominated by I_ℓ .
- I belongs to **class-c**.
From the definition of **class-c** the interval I does not intersect with I_ℓ and I_1 but intersects with I_2 . Therefore I is hop dominated by I_1 .

Hence the claim holds. $\square$

We consider the vertices of G corresponding to the set of intervals in S and such a vertex set is a hop dominating set of G of size at most four. $\square$

Lemma 3 (*). *Let G an interval graph with $\text{diam}(G)=2$. Then there exists a hop dominating set $S \subseteq V(G)$ such that $|S| \leq |U| + 2$, where U is the set of all universal vertices of G .*

Observation 2 (*). *Let G be an interval graph that admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ on $q \leq 3$ layers. Then $\text{diam}(G) \leq 3$.*

Lemma 4 (*). *Let G be an interval graph that admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_3\}$ on four layers. Then there exists a hop dominating set S such that $|S| \leq 4$.*

Lemma 5. *Let G be an interval graph that admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ on q layers. Let S be a hop dominating set of G . Then there exists a hop dominating set S' of G of size at most $|S|$ such that $|(L_i \cup L_{i+1} \cup L_{i+2} \cup L_{i+3} \cup L_{i+4}) \cap S'| \leq 8$ for all $i \in \{0, 1, \dots, q-5\}$.*

Proof. If S satisfies the desired constraints then the claim trivially holds. Therefore we assume that S does not satisfy the constraints of the lemma. We construct another hop dominating set S' from S satisfying the constraints of the lemma. Let $S' = S$ and $\mathcal{L}_i = L_i \cup L_{i+1} \cup L_{i+2} \cup L_{i+3} \cup L_{i+4}$ for each $i \in [0, q-5]$ denote the set of vertices in the layers L_j where $j \in [i, i+4]$.

We process the layers in chunks of five starting from the last layer L_{q-1} . Consider the largest $i \in [0, q-5]$ such that $|\mathcal{L}_i \cap S| > 8$. We have two cases depending on whether $i = 0$ or not.

Case: $i \neq 0$.

Let $S' = (S' \setminus \mathcal{L}_i) \cup S_i$ where $S_i = \{\beta_{i-1}, \alpha_i, \beta_i, \alpha_{i+1}, \beta_{i+1}, \alpha_{i+2}, \beta_{i+2}, \beta_{i+3}, \beta_{i+4}\}$. Notice that $|S'| \leq |S|$. We now show the following claim.

Claim. S' is a hop dominating set of G .

Proof. Consider a vertex $v \in L_j \setminus S'$ where $j \in [0, i-3] \cup [i+7, q-1]$. Since the distance between v and the vertices in $\mathcal{L}_i$ is at least three, the vertex v is hop dominated by $S' \setminus \mathcal{L}_i$. Now we consider the vertices in the layers L_j where $j \in [i-2, i+6]$ that do not belong to the set S' . If $v \in L_j$ for all $j \in [i+1, i+6]$, then the vertex β_{j-2} hop dominates v as $\beta_{j-2} \in S'$.

– $v \in L_{i-2}$.

If v was hop dominated by a vertex in $S' \setminus \mathcal{L}_i$, it continues to be hop dominated by the same vertex. Otherwise, v is hop dominated by a vertex say $w \in L_i$. Therefore exists a vertex $z \in L_{i-1}$ such that $vz \in E(G)$ and $zw \in E(G)$. Since α_i is adjacent to z and $\alpha_i \in S'$ we have that v is hop dominated by α_i .

– $v \in L_{i-1}$.

We argue this based on the following sub-cases. If $v\beta_{i-1} \notin E(G)$ then it is hop dominated by β_{i-1} because β_{i-1} is adjacent to β_{i-2} , and β_{i-2} is adjacent to v . Since $\beta_{i-1} \in S'$ and $v \notin S'$ we have that $|L_{i-1}| \geq 2$ and thus L_{i-1} is not the layer L_0 . This exhibits the existence of the vertex β_{i-2} .

Else if $v\beta_{i-1} \in E(G)$ and $v\beta_i \notin E(G)$ then v is hop dominated by β_i . Otherwise v is hop dominated by α_{i+1} , as α_{i+1} is adjacent to β_i .

– $v \in L_i$.

We argue this based on the following sub-cases. If $v\beta_i \notin E(G)$ then v is hop dominated by β_i because β_i is adjacent to β_{i-1} , and β_{i-1} is adjacent to v . Else if $v\beta_i \in E(G)$ and $v\beta_{i+1} \notin E(G)$ then v is hop dominated by β_{i+1} . Otherwise v is hop dominated by α_{i+2} , as α_{i+2} is adjacent to β_{i+1} .

□

Case: $i = 0$.

Let $S' = (S' \setminus \mathcal{L}_0) \cup S_0$ where $S_0 = \{\beta_0, \alpha_1, \beta_1, \alpha_2, \beta_2, \alpha_3, \beta_3, \beta_4\}$. Using similar arguments as before, we now show that S' is a hop dominating set.

Consider a vertex $v \in L_j \setminus S'$ where $j \in [7, q-1]$. Since the distance between v and the vertices in $\mathcal{L}_0$ is at least three, the vertex v is hop dominated by $S' \setminus \mathcal{L}_0$. Now we consider the vertices in the layers L_j where $j \in [2, 6]$. Each vertex in this set is hop dominated by β_{j-2} . The vertex in $L_0 \in S'$. Now consider a vertex

$v \in L_1 \setminus S'$. If $v\beta_1 \notin E(G)$, then v is hop dominated by β_1 . Else if $v\beta_1 \in E(G)$ and $v\beta_2 \notin E(G)$, then v is hop dominated by β_2 . Otherwise v is hop dominated by α_3 . $\square$

Theorem 2. HOP DOMINATION is polynomial time solvable on interval graphs.

Proof. Let G be an interval graph and $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ be a multi-chain ordering of G with q layers. Let S be the minimum hop dominating set for G . We design an algorithm based on the value of q .

Case: $q \leq 3$.

From Observation 2, we get that the diameter of G is at most three. Now we have the following cases. When $\text{diam}(G) = 1$ then G is clique and thus from Observation 1 we have that $S = V(G)$.

Consider $\text{diam}(G) = 2$ and let U be the set of universal vertices of G . From Observation 1 we have that $U \subseteq S$. From Lemma 3 there exists a hop dominating set of size at most $|U| + 2$. Thus any minimum hop dominating set contains at most two vertices from $V(G) \setminus U$. We go over all subsets of $V(G) \setminus U$ of size at most two and check whether it is a hop dominating set.

Consider $\text{diam}(G) = 3$. From Lemma 2 we get that any minimum hop dominating set S has $|S| \leq 4$. We go all subsets of $V(G)$ of size at most four and check whether it is a hop dominating set, and obtain our desired set.

Case: $q = 4$.

From Lemma 4, we get that the size of a minimum hop dominating set is at most four. We consider all subsets of $V(G)$ of size at most four and check whether it is a hop dominating set of G , and obtain our desired set.

Case: $q \geq 5$.

Let S be a minimum hop dominating set of G . Then by Lemma 5, there exists a hop dominating set S' of size at most S such that $|(L_i \cup L_{i+1} \cup L_{i+2} \cup L_{i+3} \cup L_{i+4}) \cap S'| \leq 8$ for each $i \in [0, q-5]$ and thus we assume $S = S'$. Notice that the vertices in L_i can be hop dominated only by the vertices from $\mathcal{M}_i$. Therefore, for each $i \in [2, q-3]$, we define a set of tuples $\mathcal{T}_i = \{(t_{i-2}, t_{i-1}, t_i, t_{i+1}, t_{i+2}) \mid t_j \in [0, 8] \text{ for each } j \in [i-2, i+2]\}$ which captures the number of solution vertices from the layer L_j for each $j \in [i-2, i+2]$.

Now for the layers L_0, L_1, L_{q-2} and L_{q-1} , we define a set of tuples $\mathcal{T}_0 = \{(t_0, t_1, t_2) \mid t_j \in [0, 8], \forall j \in [0, 2]\}$, $\mathcal{T}_1 = \{(t_0, t_1, t_2, t_3) \mid t_j \in [0, 8], \forall j \in [0, 3]\}$, $\mathcal{T}_{q-2} = \{(t_{q-4}, t_{q-3}, t_{q-2}, t_{q-1}) \mid t_j \in [0, 8], \forall j \in [q-4, q-1]\}$ and $\mathcal{T}_{q-1} = \{(t_{q-3}, t_{q-2}, t_{q-1}) \mid t_j \in [0, 8], \forall j \in [q-3, q-1]\}$ respectively.

For each tuple $T = (t_{i-2}, t_{i-1}, t_i, t_{i+1}, t_{i+2}) \in \mathcal{T}_i$ where $i \in [2, q-3]$, we have set of tuples $\mathcal{B}_T$ defined as $\mathcal{B}_T = \{(B_{i-2}, B_{i-1}, B_i, B_{i+1}, B_{i+2}) \mid B_j \subseteq L_j \text{ and } |B_j| = t_j \text{ for all } j \in [i-2, i+2]\}$.

Similarly for a tuple $T = (t_0, t_1, t_2) \in \mathcal{T}_0$ we define $\mathcal{B}_T = \{(B_0, B_1, B_2) \mid B_j \subseteq L_j \text{ and } |B_j| = t_j \text{ for all } j \in [0, 2]\}$. For a tuple $T = (t_0, t_1, t_2, t_3) \in \mathcal{T}_1$ we define $\mathcal{B}_T = \{(B_0, B_1, B_2, B_3) \mid B_j \subseteq L_j \text{ and } |B_j| = t_j \text{ for all } j \in [0, 3]\}$. For a tuple $T = (t_{q-4}, t_{q-3}, t_{q-2}, t_{q-1}) \in \mathcal{T}_{q-2}$ we define $\mathcal{B}_T = \{(B_{q-4}, B_{q-3}, B_{q-2}, B_{q-1}) \mid B_j \subseteq L_j \text{ and } |B_j| = t_j \text{ for all } j \in [q-4, q-1]\}$.

For a tuple $T = (t_{q-3}, t_{q-2}, t_{q-1}) \in \mathcal{T}_{q-1}$ we define $\mathcal{B}_T = \{(B_{q-3}, B_{q-2}, B_{q-1}) \mid B_j \subseteq L_j \text{ and } |B_j| = t_j \text{ for all } j \in [q-3, q-1]\}$.

We are now ready to describe our dynamic programming table based on the layers.

Case: For the layer L_i where $i \in [2, q-3]$.

For a fixed $T = (t_{i-2}, t_{i-1}, t_i, t_{i+1}, t_{i+2}) \in \mathcal{T}_i$ and a fixed $B = (B_{i-2}, B_{i-1}, B_i, B_{i+1}, B_{i+2}) \in \mathcal{B}_T$, the entry $h[i, T, B]$ stores the minimum size of a set $S \subseteq L_0 \cup L_1 \cup \dots \cup L_{i+2}$ such that every vertex in $L_0 \cup L_1 \cup \dots \cup L_i$ is hop dominated, and $|S \cap L_j| = t_j$ and $S \cap L_j = B_j$, for all $j \in [i-2, i+2]$.

Case: For layer L_0 .

For a fixed $T = (t_0, t_1, t_2) \in \mathcal{T}_0$ and a fixed $B = (B_0, B_1, B_2) \in \mathcal{B}_T$, the entry $h[0, T, B]$ stores the minimum size of a set $S \subseteq L_0 \cup L_1 \cup L_2$ such that every vertex in L_0 is hop dominated and $|S \cap L_j| = t_j$ and $S \cap L_j = B_j$, for all $j \in [0, 2]$.

Case: For layer L_1 .

For a fixed $T = (t_0, t_1, t_2, t_3) \in \mathcal{T}_1$ and a fixed tuple $B = (B_0, B_1, B_2, B_3) \in \mathcal{B}_T$, the entry $h[1, T, B]$ stores the minimum size of a set $S \subseteq L_0 \cup \dots \cup L_3$ such that every vertex in $L_0 \cup L_1$ is hop dominated and $|S \cap L_j| = t_j$ and $S \cap L_j = B_j$, for all $j \in [0, 3]$.

Case: For layer L_{q-2} .

For a fixed $T = (t_{q-4}, t_{q-3}, t_{q-2}, t_{q-1}) \in \mathcal{T}_{q-2}$ and a fixed $B = (B_{q-4}, B_{q-3}, B_{q-2}, B_{q-1}) \in \mathcal{B}_T$, the entry $h[q-2, T, B]$ stores the minimum size of a set $S \subseteq L_0 \cup L_1 \cup \dots \cup L_{q-2}$ such that every vertex in $L_0 \cup L_1 \cup \dots \cup L_{q-2}$ is hop dominated, and $|S \cap L_j| = t_j$ and $S \cap L_j = B_j$, for all $j \in [q-4, q-1]$.

Case: For layer L_{q-1} .

For a fixed $T = (t_{q-3}, t_{q-2}, t_{q-1}) \in \mathcal{T}_{q-1}$ and a fixed $B = (B_{q-3}, B_{q-2}, B_{q-1}) \in \mathcal{B}_T$, the entry $h[q-1, T, B]$ stores the minimum size of a set $S \subseteq L_0 \cup L_1 \cup \dots \cup L_{q-1}$ such that every vertex in $L_0 \cup L_1 \cup \dots \cup L_{q-1}$ is hop dominated, and $|S \cap L_j| = t_j$ and $S \cap L_j = B_j$, for all $j \in [q-3, q-1]$.

We now give the procedure to calculate the values of these entries.

Base case: The base cases are when $i = 0$ and $i = 1$. Let $L_0 = \{v_0\}$. For a fixed T and B ,

$$h(0, T, B) = \begin{cases} t_0 + t_1 + t_2 & \text{if } B_0 \cup B_1 \cup B_2 \text{ hop dominates } v_0 \\ \infty & \text{otherwise.} \end{cases}$$

$$h(1, T, B) = \begin{cases} t_0 + t_1 + t_2 + t_3 & \text{if } B_0 \cup B_1 \cup B_2 \cup B_3 \text{ hop dominates } L_0 \cup L_1 \\ \infty & \text{otherwise.} \end{cases}$$

We say a set X *hop dominates* Y if for each $y \in Y$ either $y \in X$ or there exists a vertex $x \in X$ such that $d(x, y) = 2$.

Recursive Step: For each $i \in [2, q-3]$, a fixed $T = (t_{i-2}, t_{i-1}, t_i, t_{i+1}, t_{i+2})$ and a fixed $B = (B_{i-2}, B_{i-1}, B_i, B_{i+1}, B_{i+2})$, we set

$$h(i, T, B) = t_{i+2} + \min_{\substack{T' \in \mathcal{T}_{i-1} \text{ respects } T \\ B' \in \mathcal{B}_{T'} \text{ respects } B}} h(i-1, T', B')$$

We say a tuple $T' = (t'_{i-3}, t'_{i-2}, t'_{i-1}, t'_i, t'_{i+1}) \in \mathcal{T}_{i-1}$ *respects* T if $t'_j = t_j$ for all $j \in [i-2, i+1]$. Similarly, we say a tuple $B' = (B'_{i-3}, B'_{i-2}, B'_{i-1}, B'_i, B'_{i+1}) \in \mathcal{B}_{T'}$ *respects* B if $B'_j = B_j$ for all $j \in [i-2, i+1]$.

For a fixed $T = (t_{q-4}, t_{q-3}, t_{q-2}, t_{q-1})$ and a fixed $B = (B_{q-4}, B_{q-3}, B_{q-2}, B_{q-1})$, we set

$$h(q-2, T, B) = \min_{\substack{T' \in \mathcal{T}_{i-1} \text{ respects } T \\ B' \in \mathcal{B}_{T'} \text{ respects } B}} h(q-3, T', B')$$

For a fixed $T = (t_{q-3}, t_{q-2}, t_{q-1})$ and $B = (B_{q-3}, B_{q-2}, B_{q-1})$, we set

$$h(q-1, T, B) = \min_{\substack{T' \in \mathcal{T}_{i-1} \text{ respects } T \\ B' \in \mathcal{B}_{T'} \text{ respects } B}} h(q-2, T', B')$$

Answer: We need a smallest sized set $S \subseteq V(G)$ such that every vertex in $V(G) \setminus S$ is hop dominated by some vertex in S . This value is attained by $\min_{T \in \mathcal{T}_{q-1}, B \in \mathcal{B}_T} h(q-1, T, B)$.

The correctness of the algorithm follows from the description of the algorithm. The number of entries in the table is $O(n^9)$ and each entry can be computed in $O(1)$ time. Due to space constraints, the detailed running time of the algorithm is presented in the full version of the paper. $\square$

5 Biconvex Bipartite Graphs

In this section, we show that HOP DOMINATION is polynomial time solvable on biconvex bipartite graphs which generalizes the result on bipartite permutation graphs [6].

Definition 3 ([9]). *A bipartite graph $G = (X, Y, E)$ satisfies the adjacency property if for every vertex $y \in Y$, the neighborhood $N(y)$ is a set of vertices that are consecutive in the ordering σ of X . A bipartite graph (X, Y, E) is biconvex if there are orderings of X (with respect to Y) and Y (with respect to X) that fulfill the adjacency property.*

It is known [10] that all connected biconvex bipartite graphs admit multi-chain ordering. Notice that since G is a bipartite graph the graph induced on each layer in a multi-chain ordering is an independent set. To prove this theorem, we will have the following lemma which is similar to Lemma 5 for biconvex bipartite graph.

Lemma 6. *Let G be an biconvex bipartite graph that admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ on q layers where $q \leq 4$. Then there is a hop dominating set S of size at most two.*

Proof. If $q = 1$ then $S = L_0$ and the claim holds trivially. Else if $q = 2$ then we have $\mathcal{L} = \{L_0, L_1\}$. Notice that the vertex in L_0 is adjacent to each vertex in L_1 . Consider the set $S = \{\beta_0, w\}$ where w is an arbitrary vertex in L_1 . Each vertex in $L_1 \setminus \{w\}$ is hop dominated by w . Else if $q = 3$, we have $\mathcal{L} = \{L_0, L_1, L_2\}$. The set $S = \{\beta_0, \beta_1\}$ is a hop dominating set of G as each vertex in $L_1 \setminus \{\beta_1\}$ (resp. L_2) is hop dominated by β_1 (resp. β_0). Otherwise $q = 4$ and we have $\mathcal{L} = \{L_0, L_1, L_2, L_3\}$. The set $S = \{\beta_0, \beta_1\}$ is a hop dominating set of G as each vertex in $L_1 \cup L_3 \setminus \{\beta_1\}$ is hop dominated by β_1 and the vertices in L_2 are hop dominated by β_0 . $\square$

Lemma 7 ($\star$). *Let G be a biconvex bipartite graph that admits a multi-chain ordering $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ on q layers. Let S be a hop dominating set of G . Then there exists a hop dominating set S' of G of size at most $|S|$ such that $|(L_i \cup L_{i+1} \cup L_{i+2} \cup L_{i+3} \cup L_{i+4}) \cap S'| \leq 7$ for all $i \in \{0, 1, \dots, q-5\}$.*

Theorem 3. HOP DOMINATION is polynomial time solvable on biconvex bipartite graphs.

Proof. Let G be a biconvex bipartite graph and $\mathcal{L} = \{L_0, L_1, \dots, L_{q-1}\}$ be a multi-chain ordering of G with q layers. If $q \leq 4$ then by Lemma 6, the size of a minimum hop dominating set is at most two. We guess all subsets of $V(G)$ of size at most two and verify whether it is a hop dominating set to obtain our desired set.

Consider the case when $q \geq 5$. From Lemma 7, there exists a hop dominating set S' of size at most $|S|$ such that every chunk of five layers contains at most seven vertices. The rest of the proof is similar to the proof of Theorem 2. $\square$

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